YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 9 OF 10 · 40 MINUTES

# Build the case 构建证据链

**No single clue proves who took the notebook: rank evidence by how many suspects it fits, set aside contaminated evidence with a checked reason.**

- I can rank evidence as strong, supporting or weak, and explain why the puddle is set aside.
- I can write a CER conclusion with a confidence level and explain how tunnel vision misleads.

**strong evidence 有力证据** **supporting evidence 支持性证据** **weak evidence 薄弱证据** **contamination 污染** **tunnel vision 隧道视野** **confirmation bias 确认偏误** **claim 主张** **reasoning 推理** **confidence 把握** **expert witness 专家证人**

## THE LESSON

| | | |
|---|---|---|
| **Do Now** | 3 | Count each suspect's ticks in the suspect tracker: nobody matches everything. Slides 2 to 3. |
| **Hook** | 2 | Week 5 Ideas Parking Board photo: students find their note; today we check them. Slide 4. |
| **Learn** | 5 | Strong fits one suspect, supporting two, weak three or four; a combination can be strong. |
| **Do** | 18 | Build A3 board (7 min); cleaning log card, set Evidence 4 aside (4 min); own CER (7 min). |
| **Think and Reflect** | 12 | Cartoon and parking board sort (3); roles and frames (4); self-assessment (4); exit ticket (1). |

**Hands-on:** No practical: groups build an A3 evidence board from evidence cards, then set aside the contaminated puddle.

**Collect or assess:** CER and photographed board: rubric strands 4 and 5, into the case file. Collect self-assessment slips and exit tickets.

## SAFETY

- No practical hazards beyond normal classroom rules.
- Soil settling bottles stay sealed on the windowsill: look only, do not open.

## WATCH OUT

- Keep cleaning log cards sealed in the envelope until Do minute 7, then give a full minute of quiet.
- Never confirm the culprit, even if every group says Dr Huang; nobody reads a model conclusion aloud.

**Language:** Chinese welcome in talk; English key words in writing. Traps: contamination/contaminated, confidence/confident.

**Next week:** Before Week 10: draft strands 1 to 5; return case files, frames. Print Week 10 set; optional prop to freezer.

## PREP FOR ONE CLASS

**before the lesson about 68 min (23 could be done by someone else); after about 108 min (some times estimated)**

### 2 OR MORE WEEKS BEFORE

- [ ] Book the Week 9 print job with reprographics (all classes) 5 min · teacher

### THE WEEK BEFORE

- [ ] Read Case Bible Section 9 and the answer key (all classes) 15 min · teacher
- [ ] Collect and count the printed worksheets, boards, frames and slips 5 min · reprographics
- [ ] Check stock; print and cut 6 card packs; cut 24 slips and 6 badges 18 min · technician
- [ ] Print 6 cleaning log cards, seal in envelope; decide roles, number badges 8 min · teacher
- [ ] Slide 4 photo; photograph soil bottles; gather Week 8 photos, notes 12 min · teacher

### THE DAY BEFORE

- [ ] Optional: print the concept cartoon and put it up (optional) 3 min · technician
- [ ] Optional, if time is short: pre-fill 'Matches' lines on the 6 card packs (optional) 15 min · teacher

### ON THE DAY, BEFORE THE LESSON

- [ ] Set out case files and resources; teacher desk in hand-out order 5 min · teacher

### AFTER THE LESSON

- [ ] Photograph every evidence board 5 min · teacher
- [ ] Clean-up: work into case files; collect them, slips, tickets; badges to box 3 min · students
- [ ] Empty soil bottles on a flowerbed or into a bucket, never the sink 10 min · technician
- [ ] Read exit tickets and case files; draft strands 1 to 5 before Week 10 90 min · teacher

## PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Evidence card pack x6 (1 A4 card sheet each, cut into 9 cards)
- Evidence board x6 (A3 landscape)
- Cleaning log card x6 (A5)
- Statement frame x6 (A3); A4 alternative x24 (optional)
- Self-assessment slip x24 (12 A4 sheets, cut in half)
- Expert witness badges x6 (1 A4 sheet, cut out)
- Teacher copies: kit card, enlarged model evidence board
- Concept cartoon x1 (A4 landscape) (optional)
- Week 5 Ideas Parking Board photo print x1 (optional)

## GET OR MAKE (NEW THIS LESSON)

- Blu Tack or glue sticks x6
- Envelope x1
- Camera or phone x1

**Have ready from earlier weeks:** Class box, Case files x24, Pencils and dictionaries x24, Role cards and frame card sets x6, Week 8 results photos and board notes, Soil bottles from Week 8 (windowsill), Week 5 Ideas Parking Board (optional)